

Francis Xiatian Zhang

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Education

Durham University, UK Oct. 2021 – May 2025

- PhD in Computer Science. Research: geometric representations for clinical video analysis, surgical workflow anticipation, and endoscopic video understanding.
- Supervisors: Dr Hubert P. H. Shum and Dr Noura Al Moubayed.

King's College London, UK Sep. 2020 – Sep. 2021

- MRes in Healthcare Technologies — Distinction.
- Project: Extracting Novel Cardiovascular Risk Factors from Routine CT Volumes.

University of Southampton, UK Sep. 2019 – Sep. 2020

- MSc in Statistics with Applications in Medicine — Distinction.

Beijing University of Chinese Medicine, China Sep. 2014 – Jul. 2019

- Bachelor of Medicine — Upper Second-Class Honours.

Research Statement

I work on robot vision, particularly perception and navigation for robotic bronchoscopy. My current research combines geometric modelling and deep learning for depth estimation, visual odometry, and airway segmentation. I am interested in how robots recover 3D structure and motion from images, and how this information can support reliable navigation in challenging environments.

Relevant Experience

Research Associate, University of Edinburgh, UK Nov. 2024 – Present

- Robot vision for bronchoscopic navigation: depth estimation, airway segmentation, visual odometry, and failure diagnosis. PI: Dr Mohsen Khadem.
- Collaborative work on continuum robot simulation and objective bronchoscopy skill assessment.

Research Assistant, Durham University, UK Oct.–Nov. 2023;
Jul.–Sep. 2024

- Edge Computing and Analytics 2.0: teaching model training, ONNX export, and deployment through Python APIs. PI: Dr Anish Jindal.

Demonstrator, Durham University, UK Nov. 2021 – Jun. 2024

- Laboratory teaching in robotics, data science, programming, and text mining; instruction with Python, MATLAB, NVIDIA Jetson, and Puzzlebot.

Research Assistant, Northumbria University, UK Apr. 2022 – Jul. 2022

- Pose-based assessment of CPR skills; multi-camera data collection with nursing students and staff. PI: Dr Merryn Constable.

Awards

2026: ICRA Best Paper Award in Medical Robotics, for [Geometry-Aware Visual Odometry for Bronchoscopic Navigation via High-Gain Observer Fusion](#) (co-author).

2020: Dean's List Award for Outstanding Achievement, University of Southampton.

Main Publications

Selected publications in robotics, computer vision, and biomedical engineering. Full list: [Google Scholar](#).

- **Francis Xiatian Zhang**, Kevin Dhaliwal and Mohsen Khadem (2026). FDRAS: Failure Diagnosis and Repair for Airway Segmentation. *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*. [paper](#) | [code](#).
- Mohammadreza Kasaei, **Francis Xiatian Zhang**, Feng Li, Farshid Alambeigi, Kevin Dhaliwal and Mohsen Khadem (2026). Geometry-Aware Visual Odometry for Bronchoscopic Navigation via High-Gain Observer Fusion. *IEEE International Conference on Robotics and Automation (ICRA)*. [arXiv](#). **Best Paper Award in Medical Robotics**.
- Zhaoxing Deng, David Hanley, **Francis Xiatian Zhang**, Kevin Dhaliwal and Mohsen Khadem (2026). A Model-Based Framework for Assessing Operator Performance in Navigational Bronchoscopy. *IEEE International Conference on Robotics and Automation (ICRA)*. [conference program](#).
- Salvatore Esposito, Matías Mattamala, Daniel Rebain, **Francis Xiatian Zhang**, Kevin Dhaliwal, Mohsen Khadem and Subramanian Ramamoorthy (2026). ROOM: A Physics-Based Continuum Robot Simulator for Photorealistic Medical Datasets Generation. *IEEE International Conference on Robotics and Automation (ICRA)*. [arXiv](#).
- **Francis Xiatian Zhang**, Hao Yao, Shengxuan Chen, Hong Zhu, Hongxiao Jia, Sisi Zheng and Hubert P. H. Shum (2026). Cross-view Multimodal Vision-Based Assessment Framework for Traditional Chinese Medicine Rehabilitation Training. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*. [paper](#) | [arXiv](#) | [code](#).
- Emile Mackute, **Francis Xiatian Zhang**, Kevin Dhaliwal and Mohsen Khadem (2025). Navigational Bronchoscopy in Critical Care via End-to-End Pose Regression. *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*. [paper](#).
- **Francis Xiatian Zhang**, Emile Mackute, Mohammadreza Kasaei, Kevin Dhaliwal, Robert Thomson and Mohsen Khadem (2025). BREA-Depth: Bronchoscopy Realistic Airway-geometric Depth Estimation. *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*. [paper](#) | [arXiv](#) | [code](#).
- **Francis Xiatian Zhang**, Shuang Chen, Xianghua Xie and Hubert P. H. Shum (2024). Depth-Aware Endoscopic Video Inpainting. *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*. [paper](#) | [arXiv](#) | [code](#).
- **Francis Xiatian Zhang**, Jingjing Deng, Robert Lieck and Hubert P. H. Shum (2024). Adaptive Anticipation: Adaptive Graph Learning From Spatial Information for Surgical Workflow Anticipation. *IEEE Transactions on Medical Robotics and Bionics*. [paper](#) | [arXiv](#) | [code](#).
- Haozheng Zhang, Edmond S. L. Ho, **Francis Xiatian Zhang**, Silvia Del Din and Hubert P. H. Shum (2024). Pose-based Tremor Type and Level Analysis for Parkinson's Disease from Video. *International Journal of Computer Assisted Radiology and Surgery*. [paper](#) | [arXiv](#).
- Haozheng Zhang, Edmond S. L. Ho, **Francis Xiatian Zhang** and Hubert P. H. Shum (2022). Pose-Based Tremor Classification for Parkinson's Disease Diagnosis from Video. *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*. [paper](#) | [arXiv](#) | [code](#).

Earlier Publications

- **Francis Xiatian Zhang**, Sisi Zheng, Hubert P. H. Shum, Haozheng Zhang, Nan Song, Mingkan Song and Hongxiao Jia (2023). Correlation-Distance Graph Learning for Treatment Response Prediction from rs-fMRI. *International Conference on Neural Information Processing (ICONIP)*, pp. 298–312. [paper](#).
- **Francis Xiatian Zhang**, Noura Al Moubayed and Hubert P. H. Shum (2022). Towards Graph Representation Learning Based Surgical Workflow Anticipation. *IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI)*, pp. 1–4. [paper](#).

Skills

- **Programming:** Python, MATLAB, and R; PyTorch and TensorFlow.
- **Computer vision:** Depth estimation, visual odometry, segmentation, pose estimation, video analysis, and graph-based modelling.
- **Medical imaging:** CT, MRI, and fMRI processing; SPM12 and FreeSurfer.
- **Statistical modelling:** Bayesian statistics, general linear models, survival analysis, and clinical trial design.

Talks and Conference Presentations

- Application of Graph Network Analysis in the Interpretation of Medical Data and Support of Clinical Decision-Making. *Beijing Integrated Medicine Committee Annual Conference, Psychiatry Workshop*, Beijing, China, November 2023.
- Correlation-Distance Graph Learning for Treatment Response Prediction from rs-fMRI. *ICONIP 2023*, Hunan, China.
- Towards Graph Representation Learning-Based Surgical Workflow Anticipation. *IEEE-EMBS BHI 2022*, Ioannina, Greece.

Service

Research Champion, National Institute for Health and Care Research (NIHR). Appointed December 2023. Contributions to inclusive trial recruitment and discussions of AI-assisted, privacy-preserving recruitment.

Peer review: IEEE International Symposium on Biomedical Imaging (ISBI) and IEEE Transactions on Neural Systems and Rehabilitation Engineering (TNSRE).

Training

- **BMVA Computer Vision Summer School 2022**, University of East Anglia, UK, July 2022. Computer vision lectures and laboratory sessions.
- **Clinical Clerkship**, Dongfang Hospital, Beijing University of Chinese Medicine, China, July 2018 – June 2019. Supervised experience in outpatient and inpatient departments.

References

Available on request.